higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE

## DIGITAL ELECTRONICS N4

## 9 APRIL 2019

**This marking guideline consists of 6 pages.**

Copyright reserved

## QUESTION 1

1.1
1.1.1 C
1.1.2 B
1.1.3 A
1.1.4 A
1.1.5 C

(5 × 1) (5)

1.2
1.2.1 C
1.2.2 A
1.2.3 E
1.2.4 B
1.2.5 F

(5 × 1) (5)

1.3
1.3.1 True
1.3.2 True
1.3.3 False
1.3.4 True
1.3.5 False

(5 × 1) (5)

1.4
1.4.1 2421 2421 2421✓
1011 1100 0100✓
=10111100,0100✓

1.4.2 8421 8421 8421✓
0101 0110 0100✓
=1010110,0100✓

1.4.3
5 6 4✓
+3 +3 +3
8 9 7✓
=10001001,0111✓

(3 × 3) (9)

**[24]**

## QUESTION 2

2.1 $19,75_{10} = 010011, 110 = 101100, 001$✓

$+ 1$

$101100, 010$✓

$15,125_{10} = 001111, 001$✓

$+ 1$

$110000, 111$✓

$101100, 010$

$110000, 111$

$011101, 001$✓

$= 35, 1_8$✓

2.2 $F, C_{16} = 1111, 1100$✓

$5, 4_8 = 101, 100$✓

$101,101$✓

$1011\overline{)111111000}$

-1011

10011

-1011✓

10000

-1011✓

10100

-1011

1001

$= 9_{10}$✓

2.3 $111100111$✓

$1101$

$111100111$✓

$000000000$

$111100111$✓

$111100111$✓

$11000101, 11011$✓

$= 305,66_8$✓

2.4 $1100, 010 = 0011, 101$✓

$1011, 001$✓

$0011, 101$

$1110, 110$✓

$0001, 001$✓

$= -0001, 001$✓

$= -1,1_8$✓

(4 × 6) **[24]**

Copyright reserved 

## QUESTION 3

3.1 $F = \overline{A\,\overline{C}\,(B + \overline{C})\,\overline{B}}$

$= \overline{A\,\overline{C}}\,(B\,\overline{B} + \overline{B}\,\overline{C})$ ✓✓

$= \overline{A} + C\,(0 + \overline{B}\,\overline{C})$ ✓

$= \overline{A}\,\overline{B}\,\overline{C} + B\,\overline{C}\,C$✓✓

$= \overline{A}\,\overline{B}\,\overline{C}$✓

3.2 LET AB = K

$= K \oplus B$✓

$= K\,\overline{B} + \overline{K}\,B$✓

$= A\,B\,\overline{B} + \overline{AB}B$✓

$= (\overline{A} + \overline{B})\,B$✓

$= \overline{A}\,B + \overline{B}\,B$✓

$= \overline{A}\,B$✓

(2 × 6) **[12]**

## QUESTION 4

4.1 4.1.1 The output of one flip-flop is used to activate the next flip-flop being fed directly onto this clock input.

4.1.2 Synchronous counters are simultaneously clocked by the same clock input pulses.

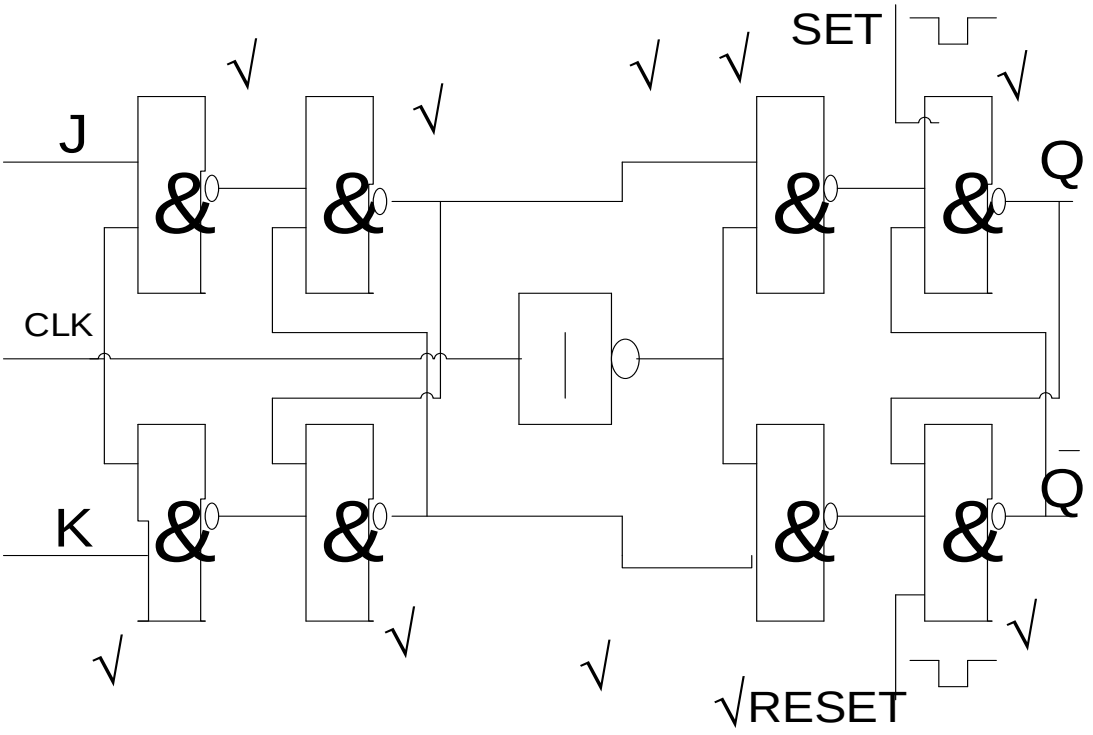
(2 × 2) (4)

4.2

| Position number | Master | Slave |
|---|---|---|
| 1 | 4.2.1 closed✓ | open |
| 2 | closed | 4.2.2 closed✓ |
| 3 | open | 4.2.3 closed✓ |
| 4 | 4.2.4 closed✓ | closed |
| 5 | 4.2.5 closed✓ | open |
| 6 | closed | 4.2.6 open✓ |

(6)

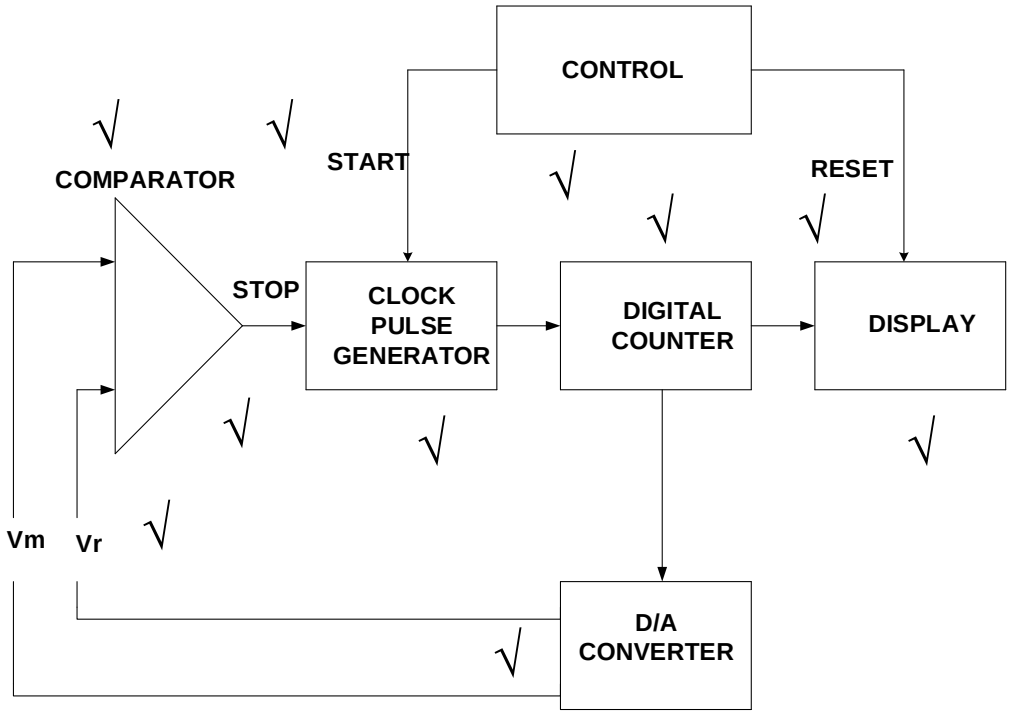
4.3




(10)
**[20]**

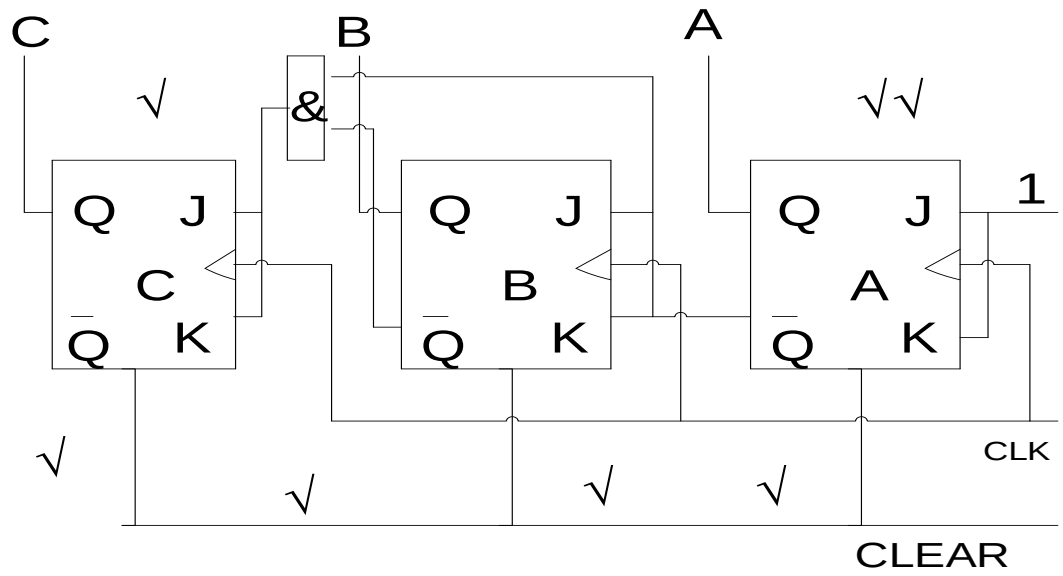
**QUESTION 5**

5.1




(10)

5.2




✓✓✓

| CLOCK | C | B | A |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 1 | 1 | 0 |
| 3 | 1 | 0 | 1 |
| 4 | 1 | 0 | 0 |
| 5 | 0 | 1 | 1 |
| 6 | 0 | 1 | 0 |
| 7 | 0 | 0 | 1 |
| 8 | 0 | 0 | 0 |

(10)
**[20]**

**TOTAL: 100**

Copyright reserved